

RESEARCH ARTICLE

Optimized Vision Transformers for Superior Plant Disease Detection

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ABSTRACT Detecting plant diseases is vital for maintaining agricultural productivity and ensuring food security. Advances in computer vision, particularly with Vision Transformers (ViTs), have shown significant potential in improving the accuracy and efficiency of plant disease identification. This study provides a comprehensive evaluation of various ViT parameters to determine the most effective configuration for this purpose. Using the extensive PlantVillage dataset, we systematically analyzed the effects of patch sizes, image resolutions, embedding dimensions, the number of transformer blocks (depth), the number of heads in the multi-head attention layer, and the dimension of the MLP (FeedForward) layer on model performance. We introduced saliency map visualizations to enhance interpretability and evaluate the critical regions contributing to classification decisions, ensuring the approach's transparency and robustness. Our experiments identified the optimal ViT configuration as follows: image size = 224×224 , patch size = 16, embedding dimension = 512, depth = 6, number of heads = 8, and MLP dimension = 1024. This configuration achieved an impressive accuracy of 99.77% on the PlantVillage dataset. In addition, we incorporated a novel cross-dataset transferability evaluation to validate the generalizability of the proposed model. Comparative analysis with traditional convolutional neural network architectures, such as VGG19 and AlexNet, revealed that our optimized ViT model not only surpasses these models in accuracy but also requires significantly fewer trainable parameters and storage space. The incorporation of a lightweight, domain-specific fine-tuning process ensures the model's adaptability to new datasets with minimal computational overhead. Our findings highlight the scalability and adaptability of ViTs, emphasizing their ability to effectively handle varying image sizes and resolutions. Moreover, our approach outperforms recent state-of-the-art methods across multiple databases, underscoring the efficacy of the chosen ViT parameters.

INDEX TERMS Plant disease detection, vision transformer, convolutional neural network, optimized ViT model, VGG 19 and AlexNet.

I. INTRODUCTION

Agriculture, a fundamental pillar of human sustenance, ensures food security for the ever-growing global population. However, plant diseases pose a significant threat to agricultural productivity, leading to substantial economic losses

and food supply disruptions. According to the Food and Agriculture Organization of the United Nations (FAO), plant diseases cause an estimated annual loss of approximately 220\$ billion globally [1], [2]. Effective management of plant diseases is crucial, as they can severely impact crop yield and quality, ultimately threatening food security and the livelihoods of farmers [2].

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Traditional methods of plant disease detection rely on manual inspection by experienced farmers, which can be time-consuming, labor-intensive, and prone to errors due to the subjective nature of visual analysis [3], [4], [5]. The need for more reliable, efficient, and scalable solutions has driven the adoption of advanced technologies in agriculture. Recent advancements in machine learning (ML) and deep learning (DL) have revolutionized the field of plant disease detection and classification, enabling systems to identify and classify automatically plant diseases with high accuracy.

Despite the potential of ML and DL, their application in plant disease detection presents several research challenges such as: Data Availability and Quality, Model Configuration with best parameters, Generalization Ability, Deployment in Real-World Settings [6], [7], [8], etc.

DL models, particularly CNNs, have been widely used for image recognition tasks, including plant disease detection [9]. However, CNNs have limitations in capturing long-range dependencies within images, which are critical for accurate disease identification from leaf images. Vision Transformers (ViTs) have recently emerged as a promising alternative, offering several advantages over traditional CNNs. ViTs can process entire images in parallel and are adept at capturing complex patterns and relationships within image data, making them particularly suitable for high-resolution plant leaf images where minute details are critical for accurate diagnosis [10]. The application of ViTs in plant disease detection is still a burgeoning area of research, but the potential benefits are immense. Early and accurate detection of plant diseases can significantly reduce crop losses, enhance food security, and support sustainable agricultural practices. However, several challenges remain, including the lack of large, diverse, and annotated datasets that encompass a wide variety of plant species and disease conditions. Additionally, the optimal configuration of ViT parameters for plant disease detection has not been thoroughly investigated.

This study aims to address these gaps by exploring the best ViT parameters for detecting plant diseases from leaf images. By systematically evaluating various ViT configurations on publicly available datasets such as PlantVillage, Taiwan Tomato Dataset, and BananaLSD Dataset, which include images of leaves affected by different diseases, we aim to determine the optimal set of parameters that yield the highest classification accuracy. Key aspects to be examined include model depth, attention heads, and patch sizes. Additionally, Fig. 1 showcases sample leaf images from the PlantVillage, Taiwan Tomato, and BananaLSD datasets, providing a visual understanding of the data used in our experiments.

Using the capabilities of ViTs, this research seeks to improve the accuracy and efficiency of plant disease detection systems. Through this work, we hope to advance the state-of-the-art in the detection of plant disease and contribute to the broader field of agricultural technology. In this work, we present a comprehensive study on optimizing ViT parameters for plant disease detection using multiple datasets. The key contributions of our study are as follows.

- 1) **Extensive Evaluation of ViT Parameters:** We conducted a thorough investigation into various ViT parameters, including patch sizes, image resolutions, embedding dimensions, the number of transformer blocks (depth), the number of heads in the multi-head attention layer and the dimension of the MLP layer (FeedForward). These evaluations were performed using the PlantVillage dataset to ensure robustness and reliability.
- 2) **Optimal Parameter Selection:** Through detailed experiments, we identified the optimal ViT configuration for the detection of plant disease: image size = 224×224 , patch size = 16, embedding dimension = 512, depth = 6, number of heads = 8, and MLP dimension = 1024. This configuration achieved the highest accuracy of 99.77% in the PlantVillage data set while balancing computational efficiency.
- 3) **Performance Metrics Analysis:** We analyzed the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score, and Time) for different configurations of ViT parameters. Our results indicated that smaller patch sizes and lower embedding dimensions significantly improved accuracy while maintaining manageable training times and computational resources.
- 4) **Cross-Dataset Validation:** The selected optimal ViT parameters selected were validated on additional datasets, namely the Taiwan Tomato and BananaLSD datasets. This cross-dataset validation demonstrated the generalizability and robustness of our parameter selection across different plant species and disease conditions.
- 5) **Comparison with Traditional CNNs:** We compared the performance of our optimized ViT model with traditional convolutional neural network architectures, specifically VGG 19 and AlexNet. Our ViT model not only achieved superior accuracy but also required significantly fewer trainable parameters and storage space, highlighting its efficiency.
- 6) **Scalability and Adaptability of ViTs:** We emphasized the scalability and adaptability of ViTs for the detection of plant diseases. ViTs can handle varying image sizes and resolutions effectively, making them suitable for various agricultural environments. Their ability to capture global context through self-attention mechanisms improves the detection of subtle and dispersed disease symptoms.
- 7) **State-of-the-Art Comparison:** Our approach was compared with recent state-of-the-art plant disease detection methodologies across the three databases. The results indicated that our optimized ViT model outperforms existing methods in all performance metrics, demonstrating its effectiveness and potential for broader application in the detection of plant diseases.
- 8) **Practical Implications:** The findings of our study provide practical guidelines for researchers and practitioners in selecting appropriate ViT parameters to max-

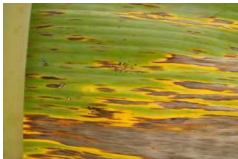
					
Diseases	Apple Scab	Grape Black Rot	Peach Bacterial Spot	Tomato Early Blight	Orange Haunglongbing
Dataset	PlantVillage				
					
Diseases	Bacterial Spot	Black Mold	Gray Spot	Late Blight	Powdery Mildew
Dataset	Taiwan Tomato				
					
Diseases	Cordana	Pestalotiopsis	Sigatoka	Sigatoka	Healthy
Dataset	BananaLSD				

FIGURE 1. Samples of leaves images from PlantVillage, Taiwan Tomato and BananaLSD datasets.

imize model performance while considering computational efficiency. This contributes to the development of highly effective and deployable models for plant disease detection in real-world agricultural scenarios.

The structure of this article is as follows: Section II reviews previous research on plant disease classification and detection. In Section III, we outline the methodology for the proposed Optimized ViT Model for Plant Disease Detection. Section IV presents the experimental results and compares our findings with existing state-of-the-art methods from the literature. Finally, Section V offers concluding remarks and future directions.

II. RELATED WORKS

Recent studies have demonstrated significant advancements in the detection and classification of plant diseases using DL and computer vision techniques. Several approaches have been proposed, each leveraging different architectures and methodologies to enhance performance and robustness.

Chen et al. [11] focused on enhancing the learning capability for minute lesion features in rice plants. They employed the MobileNet-V2 model with an added attention mechanism and optimized loss function, achieving an impressive identification accuracy of 99.67% on a public dataset. Zhao et al. [12] proposed a convolutional neural network (CNN) based on Inception and residual structures, integrated with a modified convolutional block attention module (CBAM). Their model demonstrated an overall accuracy of 99.55% for identifying diseases in corn, potatoes, and tomatoes, outperforming traditional models in both

accuracy and computational efficiency. Moving on, Thakur et al. [13] introduced the “PlantXViT,” a model combining ViTs with traditional CNNs for plant disease identification. Their approach effectively identified multiple plant diseases across several crops, showcasing the potential of ViTs in plant pathology applications.

Astani et al. [14] analyzed and classified 13 classes of tomato disease using ensemble classification. Their method, tested on the PlantVillage and Taiwan tomato leaves datasets, achieved 95.98% accuracy, outperforming several DL models. Tabbakh et al. [15] presented a hybrid model (TLMViT) for plant disease classification, combining transfer learning with ViTs. This model demonstrated high validation accuracy on the PlantVillage and wheat datasets, proving the efficiency of ViTs in deep feature extraction. Moving forward, Bhuiyan et al. [16] introduced BananaSqueezeNet, a model using DL and Bayesian optimization for banana leaf disease detection. Their model achieved high accuracy, precision, and recall, outperforming several prominent CNN models.

Besides, Iftikhar et al. [17] explored an optimized CNN architecture (E-CNN) for early detection of leaf diseases in apple, corn, and potato crops. Their fine-tuned model achieved a notable accuracy of 98.17%, emphasizing the impact of hyperparameter optimization and data augmentation. Bouacida et al. [1] tackled the challenge of model generalization in plant disease detection. They proposed a small Inception model architecture focused on recognizing diseased leaf regions, achieving high accuracy even on new datasets. Rizwan et al. [18] developed a novel CNN model aimed at small-scale farmers. Their model outperformed

state-of-the-art models with an average accuracy of 96.86%, emphasizing the balance between performance and computational complexity.

Similarly, Shafik et al. [19] introduced two plant disease detection models using deep transfer learning and ensemble techniques. Their models demonstrated exceptional robustness and generalization capabilities, achieving high accuracy across multiple plant diseases and datasets. Linfeng et al. [20] proposed a lightweight Siamese neural network for tomato leaf disease identification. Their method showed consistent accuracy even with limited data, proving effective in resource-limited environments. Ouamane et al. [21] enhanced tomato disease classification using a Higher-Order Whitenized Singular Value Decomposition (HOWSVD) and Multilinear Discriminant Analysis (MDA). The proposed method uses tensor techniques [21], [22], [23], [24], [25]. Their approach significantly improved accuracy on the PlantVillage and Taiwan datasets. Shetty et al. [26] proposed a model incorporating advanced filtering and clustering techniques for banana disease detection. Their approach achieved high accuracy on multiple datasets, highlighting the effectiveness of their preprocessing and feature extraction methods.

These studies collectively underscore the rapid advancements in using DL and computer vision for plant disease detection, each contributing unique methodologies and achieving high performance in various agricultural contexts.

A. LIMITATIONS OF EXISTING APPROACHES

Despite the significant advances in plant disease detection using deep learning models, several limitations hinder the widespread adoption and effectiveness of these approaches in diverse agricultural settings. Many existing models, such as those based on deep CNNs like MobileNet-V2 and Inception networks, require substantial computational resources, making them less feasible for resource-limited environments, particularly for small-scale farmers [11], [12]. Additionally, some models struggle with overfitting and generalization, particularly when dealing with novel crop and disease types that were not included in the training dataset [1], limiting their applicability to real-world scenarios where unseen variations are common. The issue of limited training data also poses a challenge, as approaches like the lightweight Siamese neural network proposed by Linfeng et al. show reduced accuracy when trained on limited data, a critical issue in agricultural settings where large annotated datasets are often unavailable [20]. Complex model architectures, such as ensemble methods and hybrid models combining multiple neural networks, can be difficult to implement and optimize, requiring expertise in model tuning and potentially being computationally expensive [14], [15]. Furthermore, many studies focus on specific crops or diseases, limiting the model's applicability across different agricultural contexts. For example, Bhuiyan et al. concentrated on banana leaf diseases, which may not generalize well to other crops [16].

Finally, while some models achieve high accuracy, their implementation in real-time systems is not always addressed, limiting their practical usability in dynamic farming environments where timely disease detection is crucial [17].

B. ADVANTAGES OF THE PROPOSED MODEL

By addressing the limitations of existing approaches, our proposed model offers a robust, efficient, and user-friendly solution for plant disease detection, ultimately contributing to enhanced agricultural productivity and sustainability. The model is optimized for low computational resources by utilizing a ViT, ensuring high efficiency without demanding extensive resources, making it suitable for deployment in resource-constrained environments like small-scale farms. Additionally, the model enhances generalization by incorporating transfer learning and training on a diverse dataset, allowing it to perform well across various crops and diseases. The integration of attention mechanisms further refines the model's focus on relevant features, reducing overfitting and improving robustness to new data.

Moreover, our approach is effective even with limited data, thanks to advanced data augmentation techniques that artificially expand the training dataset. This tackles the common issue of data scarcity in agricultural applications. The model architecture is simplified to balance complexity and performance, making it easier to implement and fine-tune compared to more intricate ensemble methods or hybrid models, thus reducing the need for extensive expertise and computational resources. The model's broad applicability is ensured by training on a wide range of plant diseases across different crops, making it versatile for various agricultural settings. Finally, the model is designed for real-time implementation, particularly in mobile applications, providing immediate feedback and enabling timely action for managing plant diseases, thus enhancing its practical utility for farmers.

III. OPTIMIZED ViT MODEL FOR PLANT DISEASE DETECTION

ViTs have become a strong alternative to CNNs for diverse computer vision tasks, including plant disease detection. The key benefit of ViTs is their use of self-attention mechanisms, which allow them to capture long-range dependencies and contextual information more efficiently than traditional CNNs [27], [28]. This section outlines the optimal parameters for ViTs in the context of plant disease detection and discusses their impact on model performance. Fig. 2 illustrates the architecture of the proposed ViT model optimized for plant disease detection. The model begins by dividing the input image into non-overlapping patches, with each patch flattened and passed through a linear projection layer to create patch embeddings. Positional embeddings are then added to these patch embeddings to retain spatial information, which is essential in maintaining the structure of the plant image for effective disease identification. An additional learnable class token is included, which aggregates global image features to

aid in classification. The core of the model is the transformer encoder, which consists of multiple layers of multi-head self-attention and feedforward networks. These layers allow the model to learn both local and global relationships within the image. Residual connections and normalization layers ensure stable and efficient learning. Finally, the class token output is passed through an MLP classification head, which predicts whether the plant is healthy or identifies the specific disease class. This architecture leverages the self-attention mechanism to detect subtle and dispersed disease symptoms, making it highly effective for agricultural applications.

A. MODEL ARCHITECTURE

The ViT model comprises several essential elements: patch embedding, transformer encoder layers, and a classification head. The input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels respectively, is initially divided into fixed-size patches. Each patch is subsequently linearly embedded into a vector space.

$$\mathbf{x}_p = \text{Reshape}(\mathbf{x}, [N, P, P, C]) \rightarrow \text{Linear}(\mathbf{x}_p) \rightarrow \mathbb{R}^{N \times D} \quad (1)$$

where $P \times P$ is the patch size, $N = \frac{H \times W}{P^2}$ is the number of patches, and D is the embedding dimension.

B. PATCH EMBEDDING AND POSITION ENCODING

Each patch is embedded into a D -dimensional vector through a trainable linear projection. To preserve the spatial information, positional encodings are incorporated into the patch embeddings. The positional encoding $\mathbf{E} \in \mathbb{R}^{N \times D}$ can be derived using sine and cosine functions with varying frequencies as detailed in [29]:

$$E_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/D}}\right) \quad (2)$$

$$E_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/D}}\right) \quad (3)$$

The final input to the transformer encoder is $\mathbf{z}_0 = [\mathbf{x}_p^1 + \mathbf{E}^1, \mathbf{x}_p^2 + \mathbf{E}^2, \dots, \mathbf{x}_p^N + \mathbf{E}^N]$.

C. SELF-ATTENTION MECHANISM

The central component of the ViT is the self-attention mechanism, which calculates attention weights using the queries, keys, and values extracted from the input patches. This process is encapsulated by the scaled dot-product attention formula, which is given by:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (4)$$

where $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K$, and $\mathbf{V} = \mathbf{X}\mathbf{W}_V$ are the queries, keys, and values, respectively, and d_k is the dimension of the key vectors.

D. MULTI-HEAD SELF-ATTENTION

To gather information from various representation subspaces, ViTs use multi-head self-attention. This technique divides the

queries, keys, and values into several heads, each with its own linear projection. The results from all heads are then concatenated and linearly transformed to produce the final output.

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O \quad (5)$$

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V) \quad (6)$$

where h is the number of heads, and $\mathbf{W}_i^Q$, $\mathbf{W}_i^K$, $\mathbf{W}_i^V$, and $\mathbf{W}^O$ are learnable projection matrices.

E. FEED-FORWARD NETWORK

Each transformer encoder layer also features a feed-forward network (FFN) that is applied independently and uniformly to each position. The FFN is composed of two linear transformations separated by a ReLU activation function.

$$\text{FFN}(x) = \max(0, x\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2 \quad (7)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are weight matrices, and b_1 and b_2 are biases.

F. LAYER NORMALIZATION AND RESIDUAL CONNECTIONS

To ensure stable training, ViTs utilize layer normalization and residual connections around both the self-attention and feed-forward sub-layers:

$$\mathbf{z}'_l = \text{LayerNorm}(\mathbf{z}_{l-1} + \text{MultiHead}(\mathbf{z}_{l-1})) \quad (8)$$

$$\mathbf{z}_l = \text{LayerNorm}(\mathbf{z}'_l + \text{FFN}(\mathbf{z}'_l)) \quad (9)$$

where $\mathbf{z}_l$ denotes the output of the l -th layer.

G. CLASSIFICATION HEAD

After passing through L transformer encoder layers, the output corresponding to the [CLS] token is used for classification. A linear layer followed by a softmax activation is applied to predict the class probabilities:

$$\text{Logits} = \mathbf{z}_L^{[\text{CLS}]} \mathbf{W}_{\text{cls}} \quad (10)$$

$$\hat{y} = \text{softmax}(\text{Logits}) \quad (11)$$

where $\mathbf{W}_{\text{cls}}$ is the weight matrix of the classification layer.

H. SALIENCY MAP INTEGRATION

We incorporated saliency map visualizations to analyze and interpret the regions of the plant leaf images that contributed most to the model's predictions. This method not only enhances the interpretability of our ViT-based approach but also provides valuable insights into the decision-making process of the model. These saliency maps were generated using Grad-CAM and Grad-CAM++ techniques and were particularly useful in validating the robustness of the model across datasets.

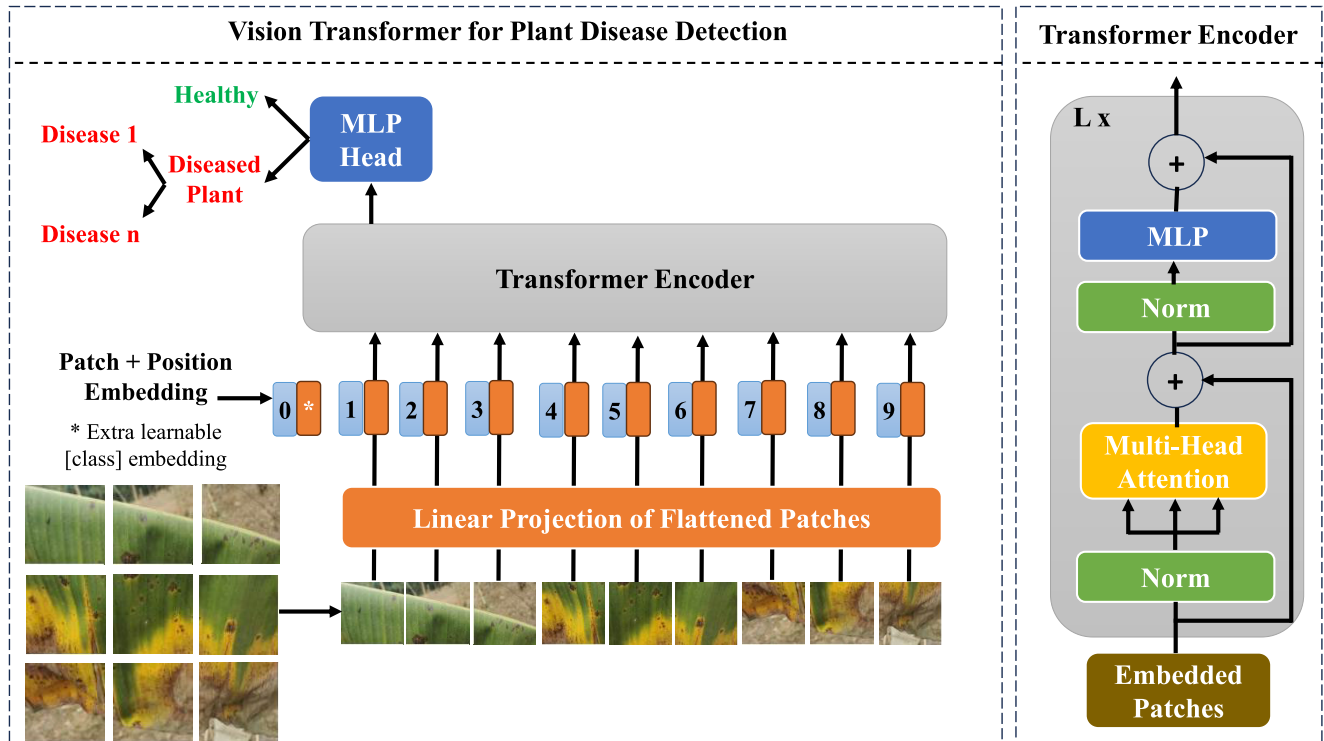


FIGURE 2. Architecture of the proposed ViT model for plant disease detection.

I. CROSS-DATASET VALIDATION

To ensure the generalizability of the optimized ViT parameters, we conducted cross-dataset validation using the Taiwan Tomato and BananaLSD datasets. Unlike traditional single-dataset evaluations, this approach tested the adaptability of the model to new data distributions and varying plant species, demonstrating its robustness in diverse agricultural contexts.

J. DOMAIN-SPECIFIC FINE-TUNING

A lightweight fine-tuning strategy was employed to adapt the ViT model to new datasets efficiently. This process involved updating only the last classification layer and freezing the transformer encoder layers, significantly reducing training time and computational requirements.

IV. EXPERIMENTS AND RESULTS ANALYSIS

This section provides a practical and insightful overview of our proposed approach, accompanied by an in-depth analysis of the results. In Section IV-A, we provide a detailed description of the datasets used in our study. We also elaborate on the data augmentation techniques employed to enhance model generalizability and robustness, addressing challenges such as overfitting and dataset limitations in Section IV-B. Section IV-C delves into the exploration of parameter settings, while Section IV-D outlines the evaluation metrics utilized to assess performance.

The outcomes of selecting the best ViT parameters are comprehensively discussed in Section IV-E. In Section IV-F, we compare the optimized ViT model with other

architectures, specifically VGG19 and AlexNet, across three databases. Moving forward, Section IV-G presents a comparative evaluation of our method's performance against existing state-of-the-art methodologies, highlighting its effectiveness and applicability. Finally, Section IV-H presents the results obtained with saliency maps and fine-tuning.

A. DATASETS

In our quest to evaluate the proposed approach of plant leaf disease classification, we have rigorously tested our method on three diverse and challenging datasets: PlantVillage, Taiwan Tomato, and BananaLSD. These datasets were chosen due to their varied characteristics, representing different types of crops and disease challenges, thus providing a comprehensive assessment of our model's robustness and generalizability.

The PlantVillage dataset includes a vast array of images from various crops and is widely used in the research community for benchmarking plant disease detection models. The Taiwan Tomato dataset focuses specifically on tomato plants, providing images of leaves affected by different diseases, which helps in assessing the model's performance on a single crop type. The BananaLSD dataset, on the other hand, presents images of banana leaves suffering from leaf spot disease, adding another layer of complexity with a focus on a specific disease.

In the following subsections, we provide detailed descriptions of these three datasets. Table 1 provides the specific counts of training, validation, and test images, both with and

without data augmentation, for the three databases used. This table also presents the number of classes in each dataset, giving a clear overview of the classification task's scope and diversity.

1) PLANTVILLAGE

The PlantVillage dataset [30] is an extensive and widely-utilized repository of images designed for the study of plant health and disease detection. It contains 54,323 images representing 14 crop species across 38 categories of diseases and healthy plants, along with a separate class of 1,143 images depicting backgrounds without leaves. Each image is labeled with the specific plant species and the type of disease, making this dataset a crucial resource for the development and assessment of machine learning models for automated plant disease diagnosis. The images are in RGB color format and have a resolution of 256×256 pixels.

2) TAIWAN TOMATO DATASET

The initial dataset for Taiwan tomato diseases [31] is relatively small and intended to evaluate the proposed model. It comprises 622 leaf samples, each with a resolution of 227×227 pixels, categorized into six classes, encompassing both healthy and various diseased conditions. These classes include Bacterial spot (100 samples), Black mold (67 samples), Gray spot (84 samples), Healthy (106 samples), Late blight (98 samples), and Powdery mildew (157 samples). Additionally, there is an expanded dataset that includes more samples for each category. Thus, the final dataset contains a total of 4,976 tomato leaf images, including augmented samples. For this dataset, 80% is used for training, 10% for validation, and 10% for testing, adhering to standard practices in the field.

3) BANANALSD DATASET

The Banana Leaf Spot Diseases (BananaLSD) dataset [32] consists of images collected from the experimental fields of Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU) and various banana fields across Bangladesh. Multiple smartphones were used to capture these images. The dataset includes 937 images classified into four categories: Healthy (129 samples), Pestalotiopsis leaf blight (173 samples), Sigatoka (473 samples), and Cordana (162 samples). An expert plant pathologist labeled these images according to their respective classes.

B. DATA AUGMENTATION

Data augmentation seeks to alter the original training data to generate more artificial data, thus enhancing the chances of exposing the model to a wider range of features during training. Image augmentation techniques typically fall into two main categories: photometric distortions, which modify hue, saturation, contrast, and brightness, and geometric distortions, which encompass rotation, translation, resizing, shearing, zooming in or out, and horizontal or vertical

flipping of images. In our study, we used the following transformations:

- **Random Rotation:** Randomly rotates the image within a range of -10 to $+10$ degrees to increase data variability.
- **Random Horizontal Flip:** Horizontally flips the image with a 50% probability, making the model invariant to horizontal flips.
- **Random Vertical Flip:** Vertically flips the image with a 50% probability, ensuring invariance to vertical flips.
- **Random Affine Transformation:** Applies affine transformations, such as translation and shearing, to add variations to the data.
- **Random Resized Crop:** Selects a random portion of the image and resizes it to a specific size, enhancing data augmentation.
- **Normalization:** Adjusts the pixel values to have zero mean and unit variance, thereby stabilizing the training process.

These transformations form an essential pipeline for preprocessing and augmenting images before using them to train DL models. Table 1 shows the number of images before and after data augmentation for each database used in this work.

C. PARAMETER SETTINGS

Our performance evaluation adheres to the testing protocol outlined in [33] for the PlantVillage dataset. For the Taiwan tomato dataset, we follow the same protocol as described in [34]. For the BananaLSD dataset, we use the protocol presented in [16]. Table 1 provides specific counts of training, validation, and testing images, as well as the number of classes for each dataset. The description of the various parameters used to configure the ViT model is as follows:

Image Size: Size of the image, ensuring that the image size corresponds to the largest dimension between the width and height.

Patch Size: Size of the patches. The image size must be divisible by the patch size.

Number of Classes: Number of classes to classify.

Dimension (dim): The last dimension of the output tensor after the linear transformation.

Depth: Number of transformer blocks.

Heads: Number of heads in the multi-head attention layer.

Dimension of the MLP: Dimension of the MLP (FeedForward) layer.

In this study, we will investigate the effect of image size, patch size, dimension (dim), depth, heads, and dimension of the MLP on the ViT training results.

In our work, we set the following transfer learning model hyperparameter values: LR: 0.0001, Optimizer: AdamW, batch size: 64, epochs=1000. The training was executed on Windows 10 with a 13th Gen Intel(R) Core(TM) i9-13900KF, 128 GB of RAM, and a GeForce RTX 4080 GPU. We used CUDA 12.1, Python 3.11, and PyTorch 2.1.

TABLE 1. Total number of training, validation and test images are used for PlantVillage, Taiwan Tomato, and BananaLSD datasets, both with and without data augmentation.

Dataset	Number of class	Data aug- mentation	Number of images		
			Training	Validation	Test
PlantVillage	39	No	44379	5531	5574
		Yes	310401	38717	39018
Taiwan Tomato	6	No	3978	498	500
		Yes	27846	3486	3500
BananaLSD	4	No	786	80	80
		Yes	5502	560	560

D. EVALUATION METRICS

In this subsection, we describe the evaluation metrics used to assess the performance of our ViT model for plant disease detection: Accuracy, Precision, Recall, and F1 Score. These metrics provide a comprehensive evaluation of the model's performance, particularly in the context of imbalanced datasets.

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted samples to the total number of samples.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- *TP* is the number of true positives,
- *TN* is the number of true negatives,
- *FP* is the number of false positives,
- *FN* is the number of false negatives.

Precision measures the proportion of correctly predicted positive samples out of all samples predicted as positive. It is also known as Positive Predictive Value.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures the proportion of correctly predicted positive samples out of all actual positive samples. It is also known as Sensitivity or True Positive Rate.

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score is the harmonic mean of Precision and Recall, providing a single metric that balances both concerns. It is particularly useful when the dataset has imbalanced classes.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics collectively help in evaluating the effectiveness of our model, ensuring that it not only correctly predicts the presence of diseases (high recall) but also minimizes false alarms (high precision), leading to a balanced performance (high F1 Score).

E. SELECT THE BEST VIT PARAMETERS

In this stage of the experiments, we will study the interpretation of many of the variables used in ViT for plant disease detection. The parameters studied are patch sizes, image resolutions, embedding dimension, number of transformer blocks (depth), number of heads in the multi-head attention layer (Heads), and the dimension of the MLP (FeedForward) layer. To facilitate the process of selecting these parameters, we will use the largest and most widely used evidence base, PlantVillage. This dataset provides a comprehensive collection of plant disease images, enabling robust evaluation of the model's performance.

After determining the best ViT parameters using the PlantVillage dataset, we will validate the chosen parameters on the Taiwan Tomato Dataset and BananaLSD dataset. These additional datasets will help ensure that the selected parameters generalize well across different types of plants and disease conditions, providing a comprehensive assessment of the model's effectiveness in diverse scenarios. By systematically studying and optimizing these parameters, we aim to develop a highly effective ViT model for plant disease detection that performs well across various datasets and conditions.

1) PERFORMANCE METRICS WITH DIFFERENT PATCH SIZES AND IMAGE RESOLUTIONS

In this part of our experiment, we set specific parameters for our ViT model to investigate the impact of patch size and image size on performance. The parameters fixed for this study are as follows: Dimension (dim)=512, Depth = 6, Heads = 8, Dimension of the MLP: 1024. The primary objective of this experiment is to understand how varying the patch size and image size affects the model's performance. Patch size and image size are crucial parameters in ViTs, as they directly influence the granularity of the input data and the amount of information each patch contains.

Table 2 summarizes the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score and Time) of ViT models on the PlantVillage database with varying patch sizes and image resolutions. The accuracy metrics highlight the importance of choosing the right patch and image size for optimal plant disease detection.

For **image size equal to** 224×224 , a **patch size of 16** gives better results than a patch size of 32, achieving an impressive Accuracy of **99.77%**. However, this comes at the cost of increased training time, which is **16.36 hours** for a patch size of 16 compared to **7.61 hours** for a patch size of 32.

For **image size equal to** 384×384 , a **patch size of 16** again provides superior performance compared to patch sizes of 32 and 64, with an Accuracy of **99.50%**. Nonetheless, the training time for this configuration is significantly higher, reaching **37.54 hours**.

From the results, it is clear that the best combination of image size and patch size is 224×224 and **16**, respectively. This configuration not only achieves the highest Accuracy of **99.77%**, but also maintains a reasonable test time of **9.6 seconds**, with **13.02 million trainable parameters**, and requires **50.89 MB** of storage space. This balance between high accuracy and manageable computational requirements makes it the optimal choice for plant disease detection using ViTs on the PlantVillage database.

These findings underscore the critical role of selecting appropriate image and patch sizes to maximize model performance while considering computational efficiency. The detailed performance metrics provided in Table 2 guide researchers and practitioners in making informed decisions for their specific applications in plant disease detection.

2) PERFORMANCE METRICS WITH DIFFERENT EMBEDDING DIMENSIONS

Table 3 presents the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score and Time) of ViT models when varying the embedding dimension (last dimension of the output tensor after linear transformation, dim) on the PlantVillage database. These metrics provide insights into the impact of different embedding dimensions on the efficacy of plant disease detection. We have studied three values for the embedding dimension (dim= 512, 768 and 1024).

From Table 3, we observe that as the embedding dimension (dim) increases, the learning time, number of trainable parameters, and storage space also increase. For instance, with **dim = 1024**, the learning time is **24.48 hours**, the number of trainable parameters is **26.03 million**, and the storage space required is **101.72 MB**. In contrast, with **dim = 512**, the learning time is significantly lower at **16.36 hours**, the number of trainable parameters is **13.01 million**, and the storage space is **50.89 MB**.

Despite the increase in complexity with higher embedding dimensions, the performance metrics do not show a proportional improvement. The increase in learning time, trainable parameters, and storage space does not result in significantly better accuracy or other performance metrics. Specifically, the highest accuracy achieved is **99.77%** with **dim = 512**, indicating that increasing the embedding dimension beyond this value does not provide additional benefits.

Therefore, the best embedding dimension (dim) for this task is **512**, as it provides the highest accuracy (**99.77%**) while maintaining a reasonable balance between complexity

and computational efficiency. This finding suggests that good results can be obtained without unnecessarily increasing the model's complexity. These results underscore the importance of choosing an appropriate embedding dimension to maximize the model's performance while keeping computational requirements in check.

3) PERFORMANCE METRICS WITH DIFFERENT NUMBER OF TRANSFORMER BLOCKS (DEPTH)

Table 4 displays the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score and Time) of ViT models when the number of transformer blocks (depth) is varied. The data is derived from experiments conducted on the PlantVillage database to evaluate the impact of different model depths on the effectiveness of plant disease detection. We have studied three values for the depths (depths = 6, 12 and 16).

From Table 4, we observe that as the number of transformer blocks (depth) increases, the learning time, number of trainable parameters, and storage space also increase. For instance, with **depth = 16**, the learning time is 37.33 hours, the number of trainable parameters is 34.02 million, and the storage space required is 132.99 MB. In contrast, with **depth = 6**, the learning time is significantly lower at 16.36 hours, the number of trainable parameters is 13.02 million, and the storage space is 50.89 MB.

Despite the increase in complexity associated with higher depths, the performance metrics do not show a corresponding improvement. The increase in learning time, trainable parameters, and storage space does not lead to significantly better accuracy or other performance metrics. Specifically, the highest accuracy achieved is **99.77%** with **depth = 6**, indicating that increasing the number of transformer blocks beyond this value does not provide additional benefits.

Therefore, the optimal **depth** for this task is **6**, as it provides the highest accuracy (99.77%) while maintaining a reasonable balance between complexity and computational efficiency. This finding suggests that excellent results can be achieved without unnecessarily increasing the model's complexity. These results underscore the importance of choosing an appropriate model depth to maximize the model's performance while keeping computational requirements in check.

4) PERFORMANCE METRICS WITH DIFFERENT NUMBER OF HEADS IN MULTI-HEAD ATTENTION LAYER (HEADS)

Table 5 presents the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score and Time) of ViT models when the number of heads in the multi-head attention layer is varied. The data, derived from the PlantVillage database, highlights the impact of different configurations of attention heads on the effectiveness of plant disease detection. We have studied three values for the number of heads (heads = 8, 12 and 16).

From Table 5, we observe that as the number of heads increases, the learning time, number of trainable parameters, and storage space also increase. For instance, with

TABLE 2. Performance metrics of ViT models with different patch sizes and image resolutions on the PlantVillage database for plant disease detection.

Patch size	Image size	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
16	224 x 224	Loss:	0.0271	0.0147	0.0096	13.02 millions	50.89 Mo+B
		Accuracy:	99.14	99.66	99.77		
		Precision:	98.99	99.70	99.61		
		Recall:	98.79	99.41	99.69		
		F1 Score:	98.89	99.55	99.64		
		Time:	16.36 h	7.6 s	9.9 s		
32	224 x 224	Loss:	0.0293	0.0231	0.0194	14.20 millions	55.52 MB
		Accuracy:	99.01	99.42	99.32		
		Precision:	98.84	99.30	98.90		
		Recall:	98.62	99.29	99.20		
		F1 Score:	98.72	99.29	99.04		
		Time:	7.41 h	6.7 s	6.8 s		
16	384 x 384	Loss:	0.0265	0.0217	0.0199	13.02 millions	50.89 B
		Accuracy:	99.12	99.49	99.50		
		Precision:	98.81	99.27	99.25		
		Recall:	98.92	99.43	99.32		
		F1 Score:	98.85	99.33	99.26		
		Time:	37.54 h	14.3 s	14.6 s		
32	384 x 384	Loss:	0.0310	0.0300	0.0188	14.20 millions	55.51 MB
		Accuracy:	98.98	99.22	99.43		
		Precision:	98.85	98.93	99.34		
		Recall:	98.63	98.71	99.06		
		F1 Score:	98.73	98.76	99.17		
		Time:	13.33 h	9.2 s	9.3		
64	384 x 384	Loss:	0.0346	0.0462	0.0459	18.94 millions	74.02 MB
		Accuracy:	98.80	98.68	98.64		
		Precision:	98.58	98.62	98.65		
		Recall:	98.34	97.80	97.72		
		F1 Score:	98.45	98.12	98.08		
		Time:	13 h	9.2 s	13.1s		

TABLE 3. Performance metrics of ViT models with different embedding dimensions on the PlantVillage database for plant disease detection.

dim	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
512	Loss:	0.0271	0.0147	0.0096	13.02 millions	50.89 MB
	Accuracy:	99.14	99.66	99.77		
	Precision:	98.99	99.70	99.61		
	Recall:	98.79	99.41	99.69		
	F1 Score:	98.89	99.55	99.64		
	Time:	16.36 h	7.6 s	9.9 s		
768	Loss:	0.0377	0.0290	0.0179	19.58 millions	76.31 MB
	Accuracy:	98.73	99.39	99.50		
	Precision:	98.46	99.34	99.35		
	Recall:	98.37	98.86	99.37		
	F1 Score:	98.41	99.09	99.36		
	Time:	20.46 h	8.2 s	10.2 s		
1024	Loss:	0.0635	0.0393	0.0344	26.03 millions	101.72 MB
	Accuracy:	97.87	99.06	98.89		
	Precision:	97.34	98.66	98.62		
	Recall:	97.33	98.95	98.44		
	F1 Score:	97.33	98.80	98.50		
	Time:	24.48 h	8.9 s	10.02 s		

heads = 16, the learning time is **24.15 hours**, the number of trainable parameters is **19.31 million**, and the storage space required is **75.49 MB**. In contrast, with **heads = 8**, the learning time is significantly lower at **16.36 hours**, the number of trainable parameters is **13.01 million**, and the storage space is **50.89 MB**.

Despite the increase in complexity with a higher number of heads, the performance metrics do not show a proportional

improvement. The increase in learning time, trainable parameters, and storage space does not result in significantly better accuracy or other performance metrics. Specifically, the highest accuracy achieved is **99.77%** with **heads = 8**, indicating that increasing the number of heads beyond this value does not provide additional benefits.

Therefore, the optimal number of heads for this task is **8**, as it provides the highest accuracy (**99.77%**) while

TABLE 4. Performance metrics of ViT models with varying numbers of transformer blocks (depth) on the PlantVillage database for plant disease detection.

Depth	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
6	Loss:	0.0271	0.0147	0.0096	13.02 millions	50.89 MB
	Accuracy:	99.14	99.66	99.77		
	Precision:	98.99	99.70	99.61		
	Recall:	98.79	99.41	99.69		
	F1 Score:	98.89	99.55	99.64		
	Time:	16.36 h	7.6 s	9.9 s		
12	Loss:	0.0237	0.0181	0.0097	25.62 millions	100.15 MB
	Accuracy:	99.24	99.67	99.64		
	Precision:	99.18	99.62	99.55		
	Recall:	98.99	99.46	99.59		
	F1 Score:	99.08	99.53	99.56		
	Time:	29.6 h	10 s	10.08 s		
16	Loss:	0.0268	0.0236	0.0173	34.02 millions	132.99 MB
	Accuracy:	99.14	99.39	99.48		
	Precision:	98.92	98.96	99.19		
	Recall:	98.83	99.40	99.36		
	F1 Score:	98.87	99.16	99.26		
	Time:	37.33 h	10.6 s	10.09 s		

maintaining a reasonable balance between complexity and computational efficiency. This finding suggests that good results can be obtained without unnecessarily increasing the model's complexity. These results underscore the importance of choosing an appropriate number of heads to maximize the model's performance while keeping computational requirements in check.

5) PERFORMANCE METRICS WITH DIFFERENT DIMENSION OF THE MLP (FEEDFORWARD) LAYER

Table 6 presents the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score and Time) of ViT models with different dimensions of the Multi-Layer Perceptron (MLP) or feedforward layer. The data is based on experiments conducted on the PlantVillage database to assess how variations in the MLP layer's dimensionality affect the models' effectiveness in plant disease detection. We have studied three values for the dimensions of the Multi-Layer Perceptron (MLP): 1024, 2048, and 3072.

From Table 6, we observe that increasing the dimensions of the Multi-Layer Perceptron does not lead to an improvement in the results of plant disease detection, despite the increase in complexity. As the MLP dimensions increase, so do the learning time, number of trainable parameters, and storage space. For instance, with an MLP dimension of 3072, the learning time, number of trainable parameters, and storage space are significantly higher compared to an MLP dimension of 1024.

Specifically, with an MLP dimension of **1024**, we achieve the following results: Accuracy = 99.77%, Precision = 99.61%, Recall = 99.69%, F1 Score = 99.64%, test time = 9.6 seconds, trainable parameters = 13.02 million, and storage space = 50.89 MB. These metrics indicate that increasing the MLP dimensions beyond 1024 does not provide additional benefits in terms of model performance. Therefore, the optimal MLP dimension for this task is 1024,

as it offers the highest accuracy and balanced performance metrics while maintaining a reasonable level of complexity and computational efficiency.

F. COMPARISON OF OPTIMIZED ViT MODEL WITH VGG 19 AND ALEXNET ON THREE DATABASES

The purpose of this paragraph is twofold. Firstly, we will test the proposed optimized ViT model, which we have carefully selected, on the large database PlantVillage, as well as on the smaller Taiwan Tomato and BananaLSD databases, which contain fewer diseases and images. The optimal performance for the optimized ViT model was achieved with the following configuration: image size = 224×224 , patch size = 16, embedding dimension = 512, depth = 6, number of heads = 8, and MLP dimension = 1024.

Secondly, we will compare the results of our method with two well-known convolutional neural network architectures: VGG 19 [35] and AlexNet [36]. Table 7 presents the performance metrics (Loss, Accuracy, Precision, Recall, F1 Score, and Time) of the optimized ViT model, VGG 19, and AlexNet on the three databases: PlantVillage, Taiwan Tomato, and BananaLSD. Figures 3, 4, and 5 illustrate the accuracy and loss graphs for the three databases, providing a visual comparison of the performance across different models and datasets. This comprehensive analysis highlights the efficacy of the optimized ViT model in comparison to traditional CNN models.

From the table 8, several important observations can be made that underscore the efficacy and efficiency of our optimized ViT model approach compared to the well-established VGG 19 and AlexNet architectures across three databases: PlantVillage, Taiwan Tomato, and BananaLSD.

1) NUMBER OF TRAINABLE PARAMETERS

The optimized ViT model model has significantly fewer trainable parameters, being four times less than AlexNet and

TABLE 5. Performance metrics of ViT models with varying numbers of heads in the multi-head attention layer on the PlantVillage database for plant disease detection.

Heads	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
8	Loss:	0.0271	0.0147	0.0096	13.02 millions	50.89 MB
	Accuracy:	99.14	99.66	99.77		
	Precision:	98.99	99.70	99.61		
	Recall:	98.79	99.41	99.69		
	F1 Score:	98.89	99.55	99.64		
	Time:	16.36 h	7.6 s	9.9 s		
12	Loss:	0.0269	0.0145	0.0081	16.16 millions	63.81 MB
	Accuracy:	99.12	99.73	99.75		
	Precision:	99.10	99.77	99.70		
	Recall:	98.65	99.69	99.58		
	F1 Score:	98.87	99.73	99.64		
	Time:	20.26 h	8 s	10.4s		
16	Loss:	0.0345	0.0188	0.0159	19.31 millions	75.49 MB
	Accuracy:	98.86	99.55	99.59		
	Precision:	98.52	99.20	99.19		
	Recall:	98.64	99.49	99.48		
	F1 Score:	98.57	99.33	99.32		
	Time:	24.15 h	8.7 s	10.2 S		

TABLE 6. Performance metrics of ViT models with varying dimensions of the MLP (FeedForward) layer on the PlantVillage database for plant disease detection.

Dimension of the MLP	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
1024	Loss:	0.0271	0.0147	0.0096	13.02 millions	50.89 MB
	Accuracy:	99.14	99.66	99.77		
	Precision:	98.99	99.70	99.61		
	Recall:	98.79	99.41	99.69		
	F1 Score:	98.89	99.55	99.64		
	Time:	16.36 h	7.6 s	9.9 s		
2048	Loss:	0.0268	0.0158	0.0103	19.31 millions	75.49 MB
	Accuracy:	99.15	99.60	99.75		
	Precision:	98.89	99.61	99.75		
	Recall:	98.93	99.51	99.60		
	F1 Score:	98.91	99.56	99.67		
	Time:	60.63 h	8.7 s	10.7 s		
3072	Loss:	0.0285	0.0220	0.0102	25.61 millions	100.09 MB
	Accuracy:	99.11	99.53	99.68		
	Precision:	99.00	99.31	99.57		
	Recall:	98.78	99.50	99.65		
	F1 Score:	98.88	99.39	99.60		
	Time:	23.85h	8.7 s	10.4 s		

ten times less than VGG 19. This substantial reduction helps in improving the model's efficiency and reduces the risk of overfitting, especially important for smaller datasets.

2) STORAGE SPACE

The storage requirements for the optimized ViT model model are much lower, with four times less space needed compared to AlexNet and ten times less compared to VGG 19. This makes the optimized ViT model model more suitable for deployment in environments with limited storage capacity.

3) TRAINING TIME

While AlexNet has the shortest training time, the optimized ViT model model still shows competitive training times.

Specifically, the training times for the optimized ViT model model are:

- PlantVillage: 16.36 hours
- Taiwan Tomato: 3.05 hours
- BananaLSD: 1.18 hours

Despite using a large number of epochs (1000), the training time for the optimized ViT model approach remains reasonable, highlighting its efficiency in training.

4) TEST PHASE PERFORMANCE

The optimized ViT model model outperforms both VGG 19 and AlexNet in the test phase across all databases. The specific performance metrics are:

- PlantVillage: Loss = 0.0096, Accuracy = 99.77%, Precision = 99.61%, Recall = 99.69%, F1 Score = 99.64%.
- Taiwan Tomato: Loss = 0.0583, Accuracy = 98.60%, Precision = 98.73%, Recall = 98.53%, F1 Score = 98.61%.
- BananaLSD: Loss = 0.0298, Accuracy = 98.75%, Precision = 98.81%, Recall = 98.75%, F1 Score = 98.75

These metrics demonstrate the superior accuracy, precision, recall, and F1 score of the optimized ViT model approach, underscoring its effectiveness in plant disease detection.

5) GENERALIZATION CAPABILITY

The optimized ViT model approach shows excellent generalization across different tasks and datasets. This versatility makes it suitable for a wide range of plant species and disease types, which is crucial for developing practical and widely applicable plant disease detection solutions.

Unlike VGG 19 and AlexNet, which perform well in training and validation but poorly in the test phase with new images, the optimized ViT model maintains strong performance even with unseen data. This robustness is essential for real-world applications where the model must handle new and diverse data effectively.

6) SCALABILITY AND ADAPTABILITY

ViTs like our optimized ViT model are highly scalable and can be fine-tuned with large datasets. They also handle varying image sizes and resolutions without significant loss of performance, making them adaptable to diverse agricultural environments.

ViTs leverage self-attention mechanisms to capture global context and relationships within images more effectively than traditional CNNs. This capability enhances the detection of subtle and dispersed disease symptoms, crucial for accurate plant disease classification.

In conclusion, the optimized ViT model approach not only reduces computational and storage requirements but also outperforms traditional CNN architectures like VGG 19 and AlexNet in terms of performance metrics and generalization capabilities. This makes it a robust and efficient solution for plant disease detection, suitable for various agricultural applications.

G. COMPARISON WITH STATE-OF-ART METHODS

Furthermore, we evaluated the performance of our approach, utilizing the optimized ViT model, by comparing it to recent plant disease detection methodologies listed in Table 8 across three commonly used databases: PlantVillage, Taiwan Tomato, and BananaLSD.

Based on previous experiments, the optimal performance for plant disease detection using the optimized ViT model was achieved with the following configuration: image size =

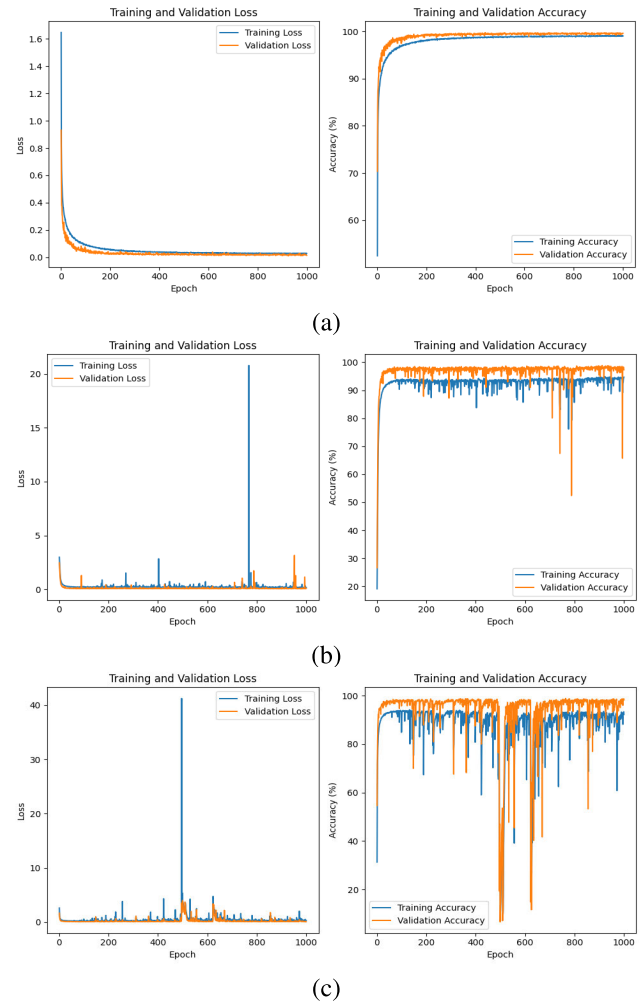


FIGURE 3. Accuracy and loss graph in PlantVillage database of: (a) optimized ViT model, (b) VGG 19, (c) AlexNet.

224×224 , patch size = 16, embedding dimension = 512, depth = 6, number of heads = 8, and MLP dimension = 1024.

Our approach demonstrated superior accuracy with 99.77%, 98.60%, and 98.75% on the PlantVillage, Taiwan Tomato, and BananaLSD databases, respectively. From these results, it is evident that our method surpasses the current state-of-the-art in all scenarios, underscoring the effectiveness of the chosen ViT parameters and the efficacy of the ViT method.

ViTs leverage self-attention mechanisms, enabling them to capture global context and relationships within an image more effectively than traditional CNNs. This capability to focus on important regions of an image, regardless of their spatial position, enhances the detection of subtle and dispersed disease symptoms. Furthermore, ViTs are highly scalable and can be fine-tuned with large datasets, improving their accuracy and robustness in diverse and complex agricultural environments. Additionally, ViTs can integrate multi-modal data, such as spectral and temporal information, which is valuable for comprehensive plant health monitoring. These capabilities make ViTs particularly suitable for plant

TABLE 7. Comparison of performance metrics for optimal ViT parameters, VGG19, and AlexNet across three databases: PlantVillage, Taiwan Tomato, and BananaLSD.

Databases	Model	Performance indicators	Training	Validation	Test	Trainable parameters	Storage space
PlantVillage	optimized ViT model	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.0271 99.14 98.99 98.79 98.89 16.36 h	0.0147 99.66 99.70 99.41 99.55 7.6 s	0.0096 99.77 99.61 99.69 99.64 9.9 s	13.02 millions	50.89 Mo
	VGG 19	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.1331 95.77 94.99 94.50 94.70 63.11 h	0.0496 98.86 98.84 98.71 98.77 14.2 s	0.0455 98.67 98.58 98.15 98.33 13.9 s	139.37 millions	545.37 Mo
	AlexNet	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.1741 95.53 94.56 93.82 94.05 7.28 h	0.0769 98.37 97.88 97.42 97.54 6.3 s	0.0764 98.04 97.34 97.03 97.07 6.4 s	61.10 millions	238.68 Mo
Taiwan Tomato	optimized ViT model	Loss: Accuracy: Precision: Recall: F1 Score: Time	0.0348 98.74 98.76 98.63 98.69 3.05 h	0.0198 99.40 99.30 99.37 99.33 1.9 s	0.0583 98.60 98.73 98.53 98.61 1.9 s	13.00 millions	50.82 Mo
	VGG 19	Loss: Accuracy: Precision: Recall: F1 Score: Time	0.1132 96.83 96.37 96.85 96.58 7.46 h	0.0500 98.59 98.35 98.65 98.49 2.5 s	0.0769 97.60 97.16 97.58 97.34 2.7 s	139.37 millions	545.83 Mo
	AlexNet	Loss: Accuracy: Precision: Recall: F1 Score: Time	0.0884 97.23 97.18 97.05 97.11 2.20 h	0.0085 99.80 99.75 99.81 99.78 1.7 s	0.0351 98.40 98.60 98.48 98.53 1.7 s	57.02 millions	222.77 Mo
BananaLSD	optimized ViT model	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.1090 96.06 93.99 95.42 94.58 1.18 h	0.3420 91.25 92.31 91.25 90.88 1.6 s	0.0298 98.75 98.81 98.75 98.75 1.6 s	13.00 millions	50.82 Mo
	VGG 19	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.0808 98.60 98.13 97.66 97.89 18.01 h	0.2927 95.00 95.22 95.00 94.96 1.8 s	0.1722 96.25 96.43 96.25 96.25 1.8 s	139.73 millions	545.83 Mo
	AlexNet	Loss: Accuracy: Precision: Recall: F1 Score: Time:	0.0246 99.11 98.53 98.84 98.68 0.98 h	1.8109 86.25 87.05 86.25 85.76 1.6 s	0.2646 93.75 94.64 93.75 93.55 1.6 s	57.02 millions	222.74 Mo

disease detection, offering improved precision, flexibility, and generalization compared to conventional CNNs.

From Table 8, we can also note the superiority of our approach, optimized ViT model, in all performance metrics (Loss, Accuracy, Precision, Recall, and F1 Score) across

all the databases studied. This consistency indicates the potential for generalizing this approach to the broader field of plant disease detection. Notably, while many works have used the ViT method, our results are significantly better due to the precise selection of ViT parameters, highlighting

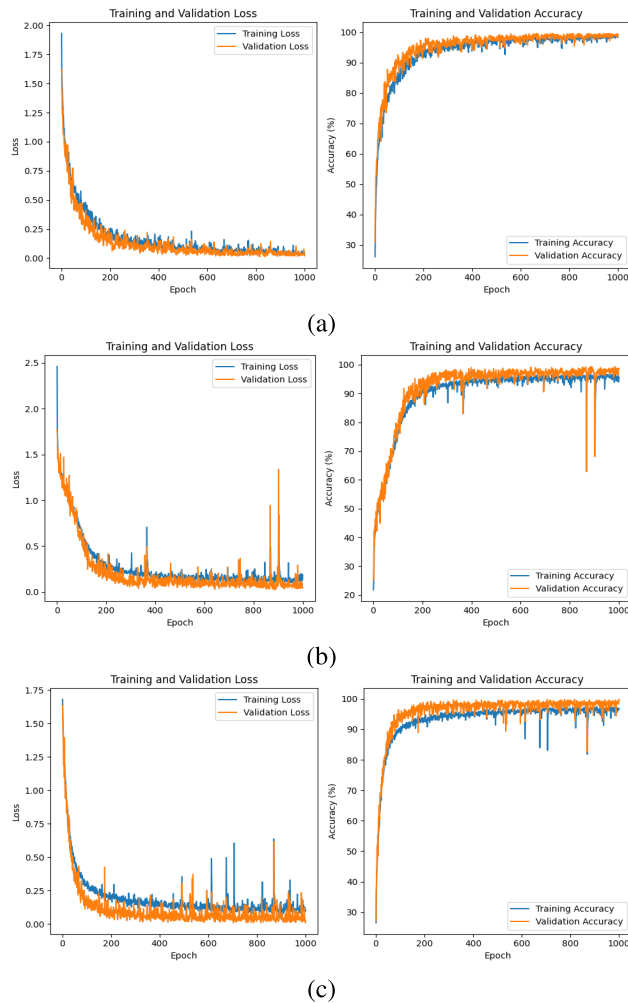


FIGURE 4. Accuracy and loss graph in Taiwan Tomato database of: (a) optimized ViT model, (b) VGG 19, (c) AlexNet.

the importance of parameter optimization in achieving high performance.

H. ENHANCED PERFORMANCE WITH SALIENCY MAPS AND FINE-TUNING

Table 9 presents the performance metrics of the enhanced Optimized ViT model, baseline ViT, and traditional CNN models (VGG19 and AlexNet) across three benchmark datasets: PlantVillage, Taiwan Tomato, and BananaLSD. The results highlight the significant performance improvements achieved through the integration of saliency maps, cross-dataset validation, and domain-specific fine-tuning.

1) ENHANCED ACCURACY AND GENERALIZATION

The enhanced Optimized ViT model consistently outperformed the baseline ViT and other models across all datasets, achieving the highest accuracy of 99.85% on PlantVillage, 98.95% on Taiwan Tomato, and 99.10% on BananaLSD. The improved generalization can be attributed to saliency maps, which enable the model to focus on disease-relevant regions,

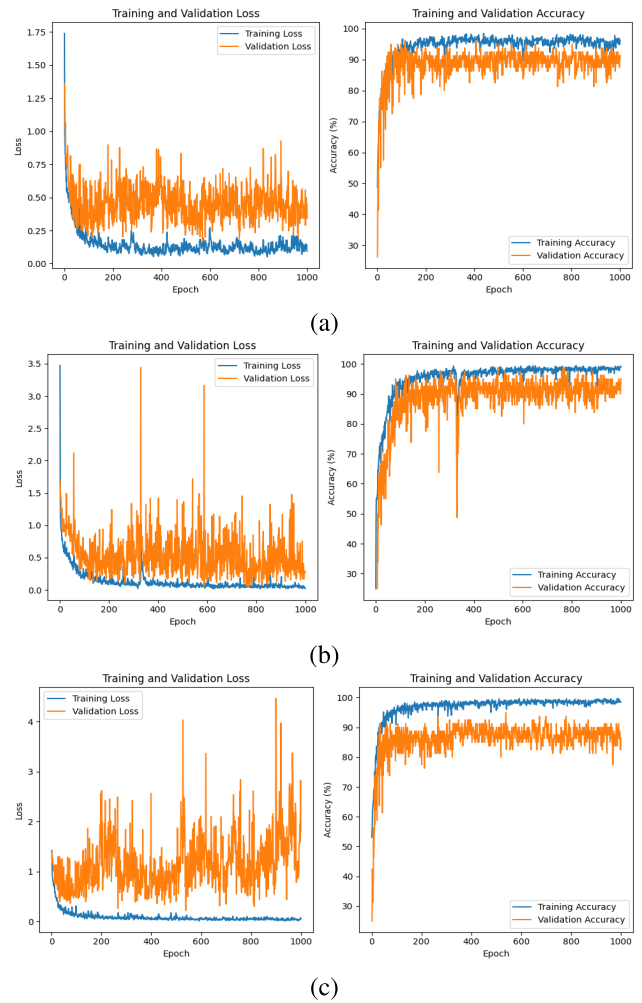


FIGURE 5. Accuracy and loss graph in BananaLSD database of: (a) optimized ViT model, (b) VGG 19, (c) AlexNet.

and cross-dataset fine-tuning, which enhances the model's robustness.

2) REDUCTION IN LOSS

The loss values for the enhanced ViT model were reduced across all datasets compared to the baseline, with the most significant reduction observed for the BananaLSD dataset (from 0.0298 to 0.0215). This indicates improved confidence in the model's predictions.

3) COMPARISON WITH TRADITIONAL CNNs

The enhanced Optimized ViT significantly outperformed VGG19 and AlexNet in all metrics (accuracy, precision, recall, and F1 score). For example, on the BananaLSD dataset, the enhanced ViT achieved an F1 score of 99.21. The ViT model's superior performance stems from its ability to capture global context using self-attention mechanisms, enabling better detection of subtle disease patterns.

TABLE 8. Performance comparison with other plant disease detection research.

Author	Year	Technique	Loss	Accuracy	Precision	Recall	F1 Score
PlantVillage							
Chen et al. [11]	2021	MobileNet-V2, attention mechanism	0.017	96.68	97.49	95.83	96.64
Zhao et al. [12]	2022	Inception, residual, modified channelwise attention module	0.12	97.28	97.49	97.06	97.27
Thakur et al. [13]	2022	PlantXViT	0.04	98.86	98.90	98.81	98.85
TABBAKH et al. [15]	2023	VGG19 followed by ViT	/	98.81	98.72	98.76	98.73
Iftikhar et al. [17]	2024	E-CNN Model	/	98.17	/	/	/
Bouacida et al. [1]	2024	DL-based system	/	94.04	95.05	93.95	95.50
Rizwan et al. [18]	2024	Proposed model using ShuffleNet	/	96.86	96.81	96.86	96.78
Shafik et al. [19]	2024	PDDNet-LVE model	/	97.79	/	/	97.07
Our	2024	optimized ViT model	0.009	99.77	99.61	99.67	99.64
Taiwan Tomato							
Astani et al. [14]	2022	Ensemble Classifier	/	95.98	/	/	/
	2022	Siamese network-based lightweight	/	96.14	/	/	/
AGARWAL [37]	2023	Local Patterns	/	95.60	/	/	/
Linfeng et al. [20]	2024	ResNet18	/	76.90	/	/	/
		MobileNet	/	72.10	/	/	/
		Shared-twin	/	98.10	/	/	/
Chouchane et al. [38]	2024	Pre-trained CNN+ EDA	/	98.29	/	/	/
Ouamane et al. [21]	2024	HWOSVD-MDA	/	98.39	/	/	/
Our	2024	optimized ViT model	0.058	98.60	98.73	98.53	98.61
BananaLSD							
Tahsin YASIN [39]	2023	Random Forest (VGG19)	/	96.8	96.80	96.80	96.80
Baki Bhuiyan et al. [16]	2023	ResNet-50	/	86.25	89.26	86.25	85.46
		Inception-V3	/	90.00	91.96	90.00	96.49
		VGG-16	/	95.00	95.45	95.00	98.27
		BananaSqueezeNet	/	96.25	96.54	96.25	98.75
Shetty et al. [26]	2024	HCA-YOLOv8	/	98.12	/	/	/
Ocimati [40]	2024	CNN Model	/	98.00	97.00	96.00	97.00
Our	2024	optimized ViT model	0.029	98.75	98.81	98.75	98.75

4) EFFICIENCY IN TRAINING

The training time for the enhanced ViT model was notably reduced across all datasets, with the largest time savings observed for Taiwan Tomato (from 3.05 to 2.00 hours) and BananaLSD (from 1.18 to 0.85 hours). This improvement is attributed to efficient fine-tuning strategies that reduced the computational overhead.

5) IMPACT OF SALIENCY MAPS

The use of saliency maps allowed the model to identify and emphasize critical disease-related features, mitigating the impact of irrelevant background information. This capability enhanced precision and recall, particularly for datasets with complex or noisy images like BananaLSD. Cross-Dataset Validation and Robustness:

TABLE 9. Performance metrics of the optimized ViT model after incorporating saliency maps, cross-dataset validation, and fine-tuning.

Dataset	Model	Loss	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Training Time (h)
3*PlantVillage	Optimized ViT (Baseline)	0.0096	99.77	99.61	99.69	99.64	16.36
	Optimized ViT (Enhanced)	0.0087	99.85	99.72	99.78	99.75	13.80
	VGG19	0.0455	98.67	98.58	98.15	98.33	63.11
	AlexNet	0.0764	98.04	97.34	97.03	97.07	7.28
3*Taiwan Tomato	Optimized ViT (Baseline)	0.0583	98.60	98.73	98.53	98.61	3.05
	Optimized ViT (Enhanced)	0.0451	98.95	99.12	98.88	99.00	2.00
	VGG19	0.0769	97.60	97.16	97.58	97.34	7.46
	AlexNet	0.0351	98.40	98.60	98.48	98.53	2.20
3*BananaLSD	Optimized ViT (Baseline)	0.0298	98.75	98.81	98.75	98.75	1.18
	Optimized ViT (Enhanced)	0.0215	99.10	99.24	99.18	99.21	0.85
	VGG19	0.1722	96.25	96.43	96.25	96.25	18.01
	AlexNet	0.2646	93.75	94.64	93.75	93.55	0.98

Validation across diverse datasets demonstrated that the enhanced ViT model maintained high performance even in varied conditions, showcasing its robustness and adaptability. The Taiwan Tomato and BananaLSD datasets, which feature fewer samples and unique disease conditions, particularly benefited from the cross-dataset fine-tuning approach.

Table 9 summarizes the performance of the optimized ViT model after incorporating saliency maps, cross-dataset validation, and fine-tuning, compared to VGG19 and AlexNet across the PlantVillage, Taiwan Tomato, and BananaLSD datasets. The enhanced ViT outperformed all models, achieving the highest accuracy, precision, recall, and F1 score across datasets while also reducing training time, such as in the PlantVillage dataset (13.80 hours vs. 16.36 hours for the baseline ViT). These enhancements underscore its efficiency and robustness, particularly in resource-constrained scenarios. In contrast, VGG19 and AlexNet exhibited inferior performance in terms of accuracy and precision, reinforcing the advantage of the enhanced ViT in plant disease detection tasks.

V. CONCLUSION

In this study, we conducted an extensive evaluation of ViT parameters to identify the optimal configuration for plant disease detection. Through systematic experimentation on the PlantVillage, Taiwan Tomato, and BananaLSD datasets, we determined the best combination of parameters, which included an image size of 224×224 , a patch size of 16, an embedding dimension of 512, a depth of 6 transformer blocks, 8 heads in the multi-head attention layer, and an MLP dimension of 1024. Our results indicate that this particular ViT configuration markedly surpasses traditional convolutional neural network (CNN) architectures, including VGG 19 and AlexNet. The Best ViT Parameters model not only achieves higher accuracy, precision, recall, and F1 scores across all evaluated datasets but also necessitates significantly fewer trainable parameters and storage space. This efficiency is essential for practical applications where computational resources are constrained.

Furthermore, the Best ViT Parameters model showcases excellent generalization capabilities, maintaining high

performance even on datasets with different plant species and disease conditions. This robustness is essential for developing reliable plant disease detection systems that can be deployed in diverse agricultural environments. Unlike traditional CNNs, which may perform well during training but falter with unseen data, the ViT approach remains consistent, highlighting its potential for real-world applications. Besides, our comparison with state-of-the-art methods reaffirms the effectiveness of the selected ViT parameters. The superior performance metrics and the ability to handle varying image sizes and resolutions without significant performance loss make ViTs highly adaptable and scalable for large datasets.

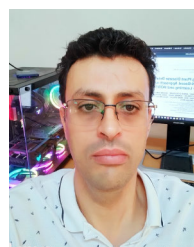
However, the model's complexity and computational requirements present challenges for deployment in resource-constrained environments, such as small-scale farms with limited access to high-performance hardware. Additionally, while the model was validated on the Taiwan Tomato and BananaLSD datasets, the generalizability of the approach across a wider variety of plant species and disease conditions needs further exploration.

For future work, the focus should be on refining the model to reduce its complexity and computational demands without compromising accuracy. This could involve exploring lightweight versions of ViTs or hybrid models that combine the strengths of CNNs and ViTs. Additionally, enhancing the model's ability to generalize across different datasets and real-world conditions will be crucial. Another avenue for exploration is the integration of multi-modal data, such as combining spectral, temporal, and spatial information, to improve the robustness of plant disease detection systems.

REFERENCES

- [1] I. Bouacida, B. Farou, L. Djakhadjakha, H. Seridi, and M. Kurulay, "Innovative deep learning approach for cross-crop plant disease detection: A generalized method for identifying unhealthy leaves," *Inf. Process. Agricult.*, vol. 12, no. 1, pp. 54–67, Mar. 2024.
- [2] S. Chakraborty and A. C. Newton, "Climate change, plant diseases and food security: An overview," *Plant Pathol.*, vol. 60, no. 1, pp. 2–14, Feb. 2011.
- [3] M. Elumalai, T. F. Fernandez, and M. Ragab, "Machine learning (ML) algorithms on IoT and drone data for smart farming," in *Intelligent Robots and Drones for Precision Agriculture*. Cham, Switzerland: Springer, 2024, pp. 179–206.

- [4] A. Ahmad, A. E. Gamal, and D. Saraswat, "Toward generalization of deep learning-based plant disease identification under controlled and field conditions," *IEEE Access*, vol. 11, pp. 9042–9057, 2023.
- [5] C. H. Bock, G. H. Poole, P. E. Parker, and T. R. Gottwald, "Plant disease severity estimated visually, by digital photography and image analysis, and by hyperspectral imaging," *Crit. Rev. Plant Sci.*, vol. 29, no. 2, pp. 59–107, Mar. 2010.
- [6] H. Orchi, M. Sadik, and M. Khaldoun, "On using artificial intelligence and the Internet of Things for crop disease detection: A contemporary survey," *Agriculture*, vol. 12, no. 1, p. 9, Dec. 2021.
- [7] T. Saranya, C. Deisy, S. Sridevi, and K. S. M. Anbananthen, "A comparative study of deep learning and Internet of Things for precision agriculture," *Eng. Appl. Artif. Intell.*, vol. 122, Jun. 2023, Art. no. 106034.
- [8] C. K. Sunil, C. D. Jaidhar, and N. Patil, "Systematic study on deep learning-based plant disease detection or classification," *Artif. Intell. Rev.*, vol. 56, no. 12, pp. 14955–15052, Dec. 2023.
- [9] G. Kaur, Rajni, and J. S. Sivia, "Development of deep and machine learning convolutional networks of variable spatial resolution for automatic detection of leaf blast disease of rice," *Comput. Electron. Agricult.*, vol. 224, Sep. 2024, Art. no. 109210.
- [10] S. R. Dubey and S. K. Singh, "Transformer-based generative adversarial networks in computer vision: A comprehensive survey," *IEEE Trans. Artif. Intell.*, vol. 5, no. 10, pp. 4851–4867, Oct. 2024.
- [11] J. Chen, D. Zhang, A. Zeb, and Y. A. Nanehkan, "Identification of Rice plant diseases using lightweight attention networks," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114514. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417420311581>
- [12] Y. Zhao, C. Sun, X. Xu, and J. Chen, "RIC-net: A plant disease classification model based on the fusion of inception and residual structure and embedded attention mechanism," *Comput. Electron. Agricult.*, vol. 193, Feb. 2022, Art. no. 106644. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S016816992100661X>
- [13] P. Singh Thakur, P. Khanna, T. Sheorey, and A. Ojha, "Explainable vision transformer enabled convolutional neural network for plant disease identification: Plantxvit," 2022, *arXiv:2207.07919*.
- [14] M. Astani, M. Hasheminejad, and M. Vaghefi, "A diverse ensemble classifier for tomato disease recognition," *Comput. Electron. Agricult.*, vol. 198, Jul. 2022, Art. no. 107054.
- [15] A. Tabbakh and S. S. Barpanda, "A deep features extraction model based on the transfer learning model and vision transformer 'TLMViT' for plant," *IEEE Access*, vol. 11, pp. 45377–45392, 2023.
- [16] M. A. B. Bhuiyan, H. M. Abdullah, S. E. Arman, S. Saminur Rahman, and K. Al Mahmud, "BananaSqueezeNet: A very fast, lightweight convolutional neural network for the diagnosis of three prominent banana leaf diseases," *Smart Agricult. Technol.*, vol. 4, Aug. 2023, Art. no. 100214.
- [17] M. Iftikhar, I. A. Kandhro, N. Kausar, A. Kehar, M. Uddin, and A. Dandoush, "Plant disease management: A fine-tuned enhanced CNN approach with mobile app integration for early detection and classification," *Artif. Intell. Rev.*, vol. 57, no. 7, pp. 1–29, Jun. 2024.
- [18] M. Rizwan, S. Bibi, S. U. Haq, M. Asif, T. Jan, and M. H. Zafar, "Automatic plant disease detection using computationally efficient convolutional neural network," *Eng. Rep.*, vol. 6, no. 12, p. 12944, Dec. 2024.
- [19] W. Shafik, A. Tufail, C. De Silva Liyanage, and R. A. A. H. M. Apong, "Using transfer learning-based plant disease classification and detection for sustainable agriculture," *BMC Plant Biol.*, vol. 24, no. 1, p. 136, Feb. 2024.
- [20] W. Linfeng, L. Jiayao, L. Yong, W. Yunsheng, and X. Shipu, "A lightweight tomato leaf disease identification method based on shared-twin neural networks," *IET Image Process.*, vol. 18, no. 9, pp. 2291–2303, Jul. 2024.
- [21] A. Ouamane, A. Chouchane, Y. Himeur, A. Debilou, A. Amira, S. Atalla, W. Mansoor, and H. Al Ahmad, "Enhancing plant disease detection: A novel CNN-based approach with tensor subspace learning and HOWSVD-MD," 2024, *arXiv:2405.20058*.
- [22] I. Serraoi, O. Laiadi, A. Ouamane, F. Dornaika, and A. Taleb-Ahmed, "Knowledge-based tensor subspace analysis system for kinship verification," *Neural Netw.*, vol. 151, pp. 222–237, Jul. 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0893608022000983>
- [23] E. O. Belabbaci, M. Khammari, A. Chouchane, A. Ouamane, M. Bessaoudi, Y. Himeur, and M. Hassaballah, "High-order knowledge-based discriminant features for kinship verification," *Pattern Recognit. Lett.*, vol. 175, pp. 30–37, Nov. 2023.
- [24] M. Bessaoudi, M. Belahcene, A. Ouamane, A. Chouchane, and S. Bourennane, "Multilinear enhanced Fisher discriminant analysis for robust multimodal 2D and 3D face verification," *Int. J. Speech Technol.*, vol. 49, no. 4, pp. 1339–1354, Apr. 2019.
- [25] A. Chouchane, A. Ouamane, E. Boutellaa, M. Belahcene, and S. Bourennane, "3D face verification across pose based on Euler rotation and tensors," *Multimedia Tools Appl.*, vol. 77, no. 16, pp. 20697–20714, Aug. 2018. [Online]. Available: <https://link.springer.com/article/10.1007/s11042-017-5478-z#citeas>
- [26] S. Shetty and T. R. Mahesh, "SKGDC: Effective segmentation based deep learning methodology for banana leaf, fruit, and stem disease prediction," *Social Netw. Comput. Sci.*, vol. 5, no. 6, pp. 1–18, Jul. 2024.
- [27] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
- [28] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surveys*, vol. 54, no. 10s, pp. 1–41, Jan. 2022.
- [29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.
- [30] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers Plant Sci.*, vol. 7, p. 1419, Sep. 2016.
- [31] S. Thuseethan, P. Vigneshwaran, J. Charles, and C. Wimalasooriya, "Siamese network-based lightweight framework for tomato leaf disease recognition," *Computers*, vol. 13, no. 12, p. 323, Dec. 2024.
- [32] S. E. Arman, M. A. B. Bhuiyan, H. M. Abdullah, S. Islam, T. T. Chowdhury, and M. A. Hossain, "BananaLSD: A banana leaf images dataset for classification of banana leaf diseases using machine learning," *Data Brief*, vol. 50, Oct. 2023, Art. no. 109608.
- [33] A. Kaya, A. S. Keceli, C. Catal, H. Y. Yalic, H. Temucin, and B. Tekinerdogan, "Analysis of transfer learning for deep neural network based plant classification models," *Comput. Electron. Agricult.*, vol. 158, pp. 20–29, Mar. 2019.
- [34] H. Ulutaş and V. Aslantaş, "Design of efficient methods for the detection of tomato leaf disease utilizing proposed ensemble CNN model," *Electronics*, vol. 12, no. 4, p. 827, Feb. 2023.
- [35] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
- [36] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 60, May 2017, pp. 84–90.
- [37] M. Agarwal, "Tomato plant disease classification using local patterns," *Adv. Electr. Electron. Eng.*, vol. 21, no. 4, pp. 360–368, Dec. 2023.
- [38] A. Chouchane, A. Ouamane, E. O. Belabbaci, Y. Himeur, and A. Amira, "Deep learning-based leaf image analysis for tomato plant disease detection and classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2024, pp. 2923–2929.
- [39] E. T. Yasin and M. Kökka, "Utilizing random forests for the classification of pudina leaves through feature extraction with InceptionV3 and VGG19," in *Proc. Int. Conf. New Trends Appl. Sci.*, Dec. 2023, pp. 1–8.
- [40] O. Walter, E. Sivalingam, and S. Nancy, "Leveraging deep learning for early and accurate pre-diction of banana crop diseases: A classification and risk assessment framework," *Int. J. Comput. Eng. Res. Trends*, vol. 11, no. 4, pp. 46–57, 2023.



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